PL UNIT 2

Q1.Write a program that demonstrates the order of execution of operators in expressions.

```
#include <stdio.h>

int main()
{
    int a, b, c;

    a = 10 + 5 * 2;
    b = (10 + 5) * 2;
    c = 20 / 5 * 2;

    printf("10 + 5 * 2 = %d\n", a);
    printf("(10 + 5) * 2 = %d\n", b);
    printf("20 / 5 * 2 = %d\n", c);

    return 0;
}
```

OUTPUT-

```
10 + 5 * 2 = 20
(10 + 5) * 2 = 30
20 / 5 * 2 = 8
```

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Q2.Write a program to check whether a given number is even or odd.

```
#include <stdio.h>

int main()
{
    int num;

    printf("Enter a number: ");
    scanf("%d", &num);

    if (num % 2 == 0)
    {
        printf("%d is an even number.\n", num);
    }
    else
    {
        printf("%d is an odd number.\n", num);
    }

    return 0;
}
```

OUTPUT -

```
Enter a number: 10
10 is an even number.
```

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Q3.Write a program to determine whether a given number is positive, negative, or zero.

```
#include <stdio.h>

int main()
{
    int num;

    printf("Enter a number: ");
    scanf("%d", &num);

    if (num > 0)
    {
        printf("The number is positive.\n");
    }
    else if (num < 0)
    {
        printf("The number is negative.\n");
    }
    else
    {
        printf("The number is zero.\n");
    }

    return 0;
}
```

OUTPUT-

```
Enter a number: 10
The number is positive.
```

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Q4.Write a program using the switch-case statement to create a menu-driven calculator that repeatedly
performs arithmetic operations until the user exits the program.

```
#include <stdio.h>

int main()
{
    int choice;
    float a, b;

    do
    {
        printf("\n--- MENU-DRIVEN CALCULATOR ---\n");
        printf("1. Addition\n");
        printf("2. Subtraction\n");
        printf("3. Multiplication\n");
        printf("4. Division\n");
        printf("5. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);

        if (choice >= 1 && choice <= 4)
        {
            printf("Enter two numbers: ");
            scanf("%f %f", &a, &b);
        }

        switch (choice)
        {
            case 1:
                printf("Result = %.2f\n", a + b);
                break;

            case 2:
                printf("Result = %.2f\n", a - b);
                break;

            case 3:
                printf("Result = %.2f\n", a * b);
```

```
            break;

        case 4:
            if (b != 0)
                printf("Result = %.2f\n", a / b);
            else
                printf("Division by zero is not allowed.\n");
            break;

        case 5:
            printf("Exiting the calculator.\n");
            break;

        default:
            printf("Invalid choice! Please try again.\n");
      }

   } while (choice != 5);

   return 0;
}
```

OUTPUT-

```
--- MENU-DRIVEN CALCULATOR ---
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice: 1
Enter two numbers: 10 5
Result = 15.00

--- MENU-DRIVEN CALCULATOR ---
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice: 3
Enter two numbers: 4 5
Result = 20.00

--- MENU-DRIVEN CALCULATOR ---
1. Addition
2. Subtraction
```

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3. Multiplication
4. Division
5. Exit
Enter your choice: 5
Exiting the calculator.

Q5.Write a program to generate the multiplication table of any number using while loop, do while loop and for loop.

```
#include <stdio.h>

int main()
{
    int num, i;

    printf("Enter a number: ");
    scanf("%d", &num);

    // Using for loop
    printf("\nMultiplication table using for loop:\n");
    for (i = 1; i <= 10; i++)
    {
        printf("%d x %d = %d\n", num, i, num * i);
    }

    // Using while loop
    printf("\nMultiplication table using while loop:\n");
    i = 1;
    while (i <= 10)
    {
        printf("%d x %d = %d\n", num, i, num * i);
        i++;
    }

    // Using do-while loop
    printf("\nMultiplication table using do-while loop:\n");
    i = 1;
    do
    {
        printf("%d x %d = %d\n", num, i, num * i);
        i++;
    } while (i <= 10);

    return 0;
}
```

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OUTPUT-

Enter a number: 5

Multiplication table using for loop:
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50

Multiplication table using while loop:
5 x 1 = 5
...
5 x 10 = 50

Multiplication table using do-while loop:
5 x 1 = 5
...
5 x 10 = 50